# PMGSY Mini Project — PySpark Analysis of Road & Bridge Data

This mini project analyzes 2,292 records across 14 columns describing road and bridge projects under multiple PMGSY phases (PMGSY-I, II, III, RCPLWEA, PM-JANMAN). The objective: clean the data with PySpark, compute completion metrics, and visualize regional and scheme-level performance to guide resource prioritization.

# Dataset Summary & Scope

Rows: 2,292 | Columns: 14. Scope includes state/district identifiers, PMGSY scheme phase, counts of roads and bridges sanctioned/completed/pending, lengths (km), sanctioned costs and expenditures (₹ Lakhs), and computed completion percentages. The data yields a clear regional overview of sanctioned vs completed infrastructure projects.

## Key Uses

Assess development stage by state/scheme, evaluate cost utilization, and identify lagging regions needing intervention.

## Primary Metrics

NO_OF_ROAD_WORK_SANCTIONED, NO_OF_ROAD_WORKS_COMPLETED, LENGTH_OF_ROAD_WORK_SANCTIONED_KM, COST_OF_WORKS_SANCTIONED_LAKHS, EXPENDITURE_OCCURED_LAKHS, Completion_Percentage, Bridge_Completion_Percentage.

# Data Cleaning & Preparation (PySpark)

Workflow: Read CSV with header & inferred schema, drop duplicates, count nulls per column, fill numeric nulls with column means, fill string nulls with 'Unknown', and standardize column names (replace spaces with underscores). This ensures reliable aggregations and visualizations.

### Read & Inspect

spark.read CSV → df.show(), df.printSchema(), df.count()

### De-duplicate & Null Handling

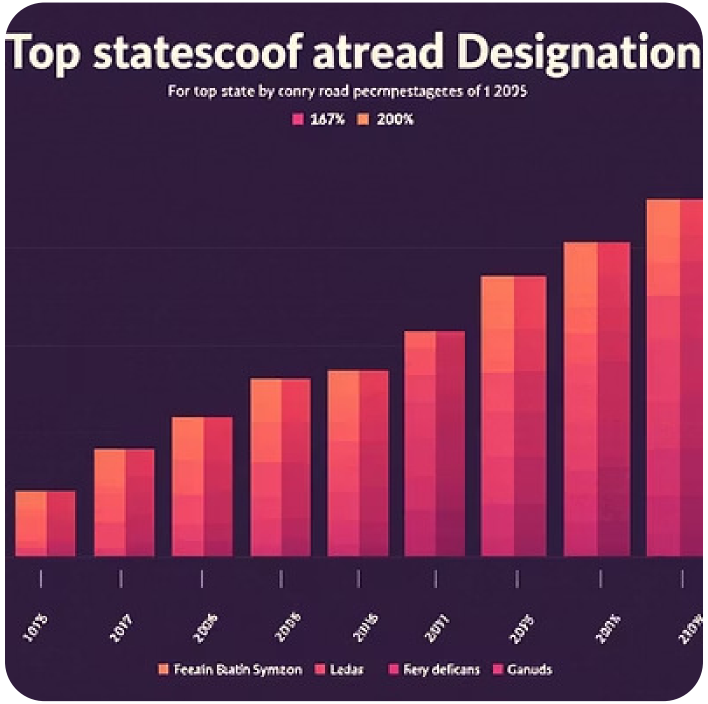
df.dropDuplicates(); count nulls; fill numeric means and string 'Unknown'

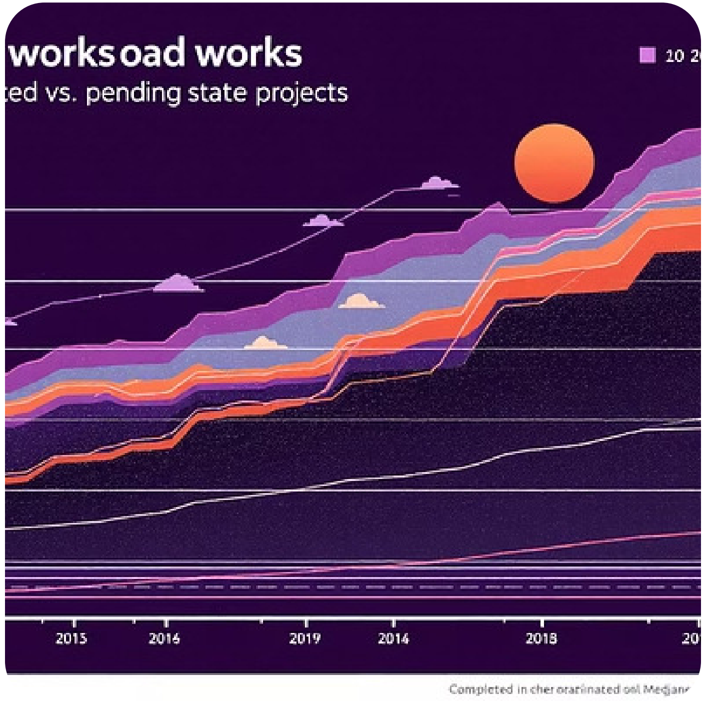
### Rename & Compute

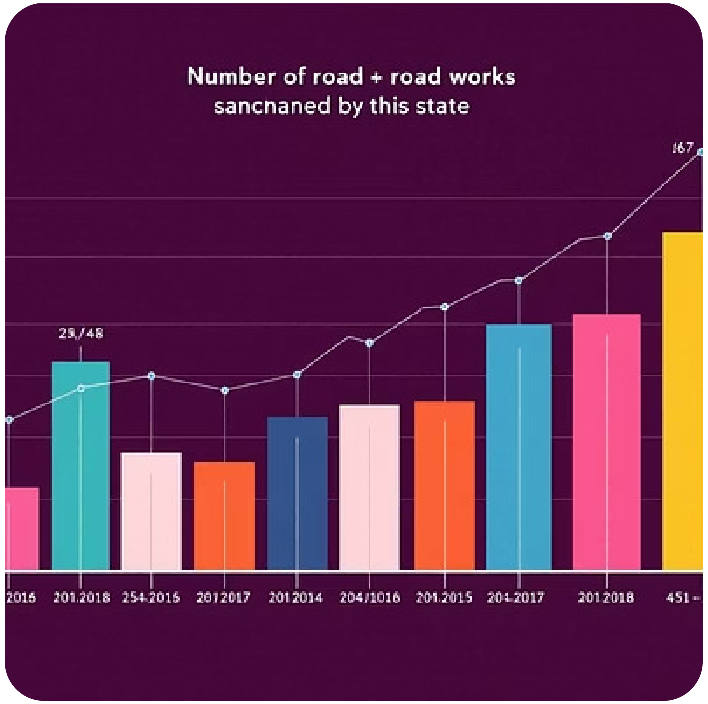
Normalize column names and compute Completion_Percentage and Bridge_Completion_Percentage safely with try_divide/round.

# Exploratory Visualizations — State Performance

Top visuals: bar charts for Top 10 states by average completion %, stacked bars for completed vs balance, and bar for total number of road works sanctioned. These reveal regional concentration, execution success, and where backlog remains.



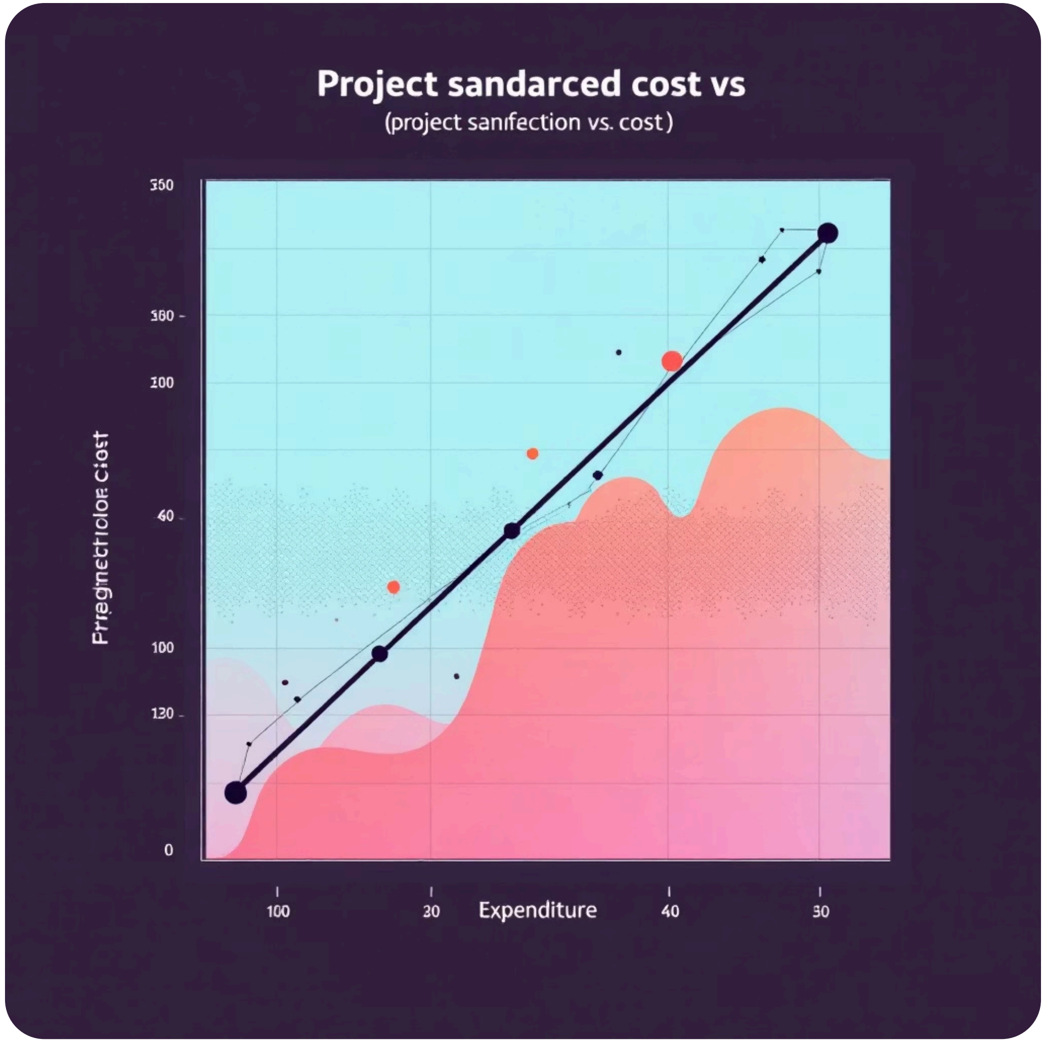




The analysis highlights states like Andhra Pradesh, Bihar, and Odisha among top performers for completion rates; Uttar Pradesh, Madhya Pradesh, Rajasthan, Bihar and Odisha lead in sanctioned counts.

# Cost vs Expenditure — Financial Discipline

Scatter plots of sanctioned cost (Lakhs) vs expenditure reveal a strong positive correlation: expenditure generally tracks sanctioned cost, indicating disciplined fund utilization. Outliers and deviations help flag projects with underspend or overspend for fiscal review.




## Key Observations

- Strong positive correlation between sanctioned cost and expenditure.
- Most projects closely track sanctioned budgets; exceptions warrant audit or field investigation.

# Length Sanctioned vs Completed — Implementation Progress

Plots of length sanctioned (KM) vs length completed (KM) show a strong linear relationship; many points align near the y=x diagonal, indicating that sanctioned road lengths are often realized. This confirms effective project execution for a large portion of records.

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### Interpretation

High alignment with diagonal → near-complete implementation. Deviations indicate partial progress or recently sanctioned projects.

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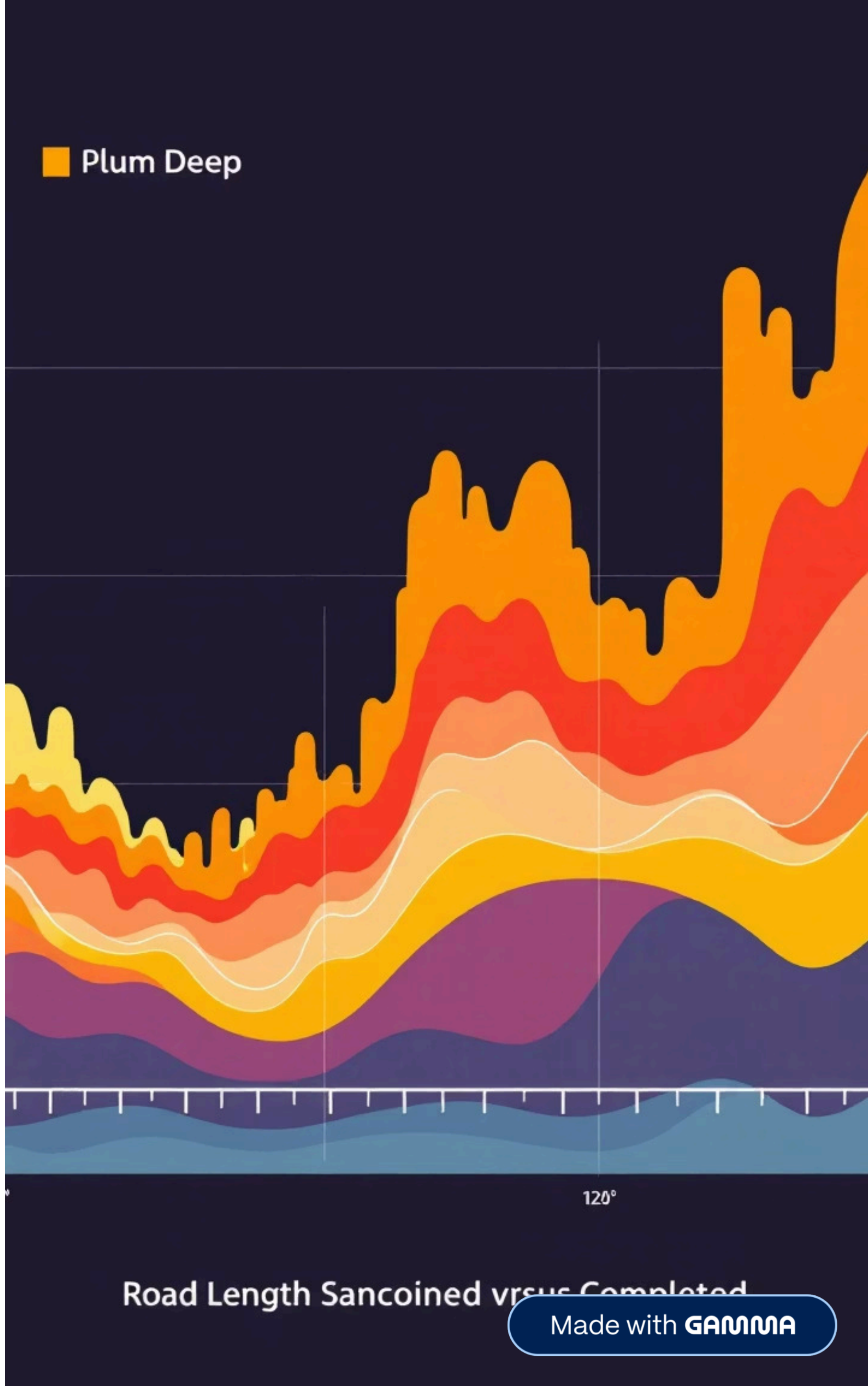
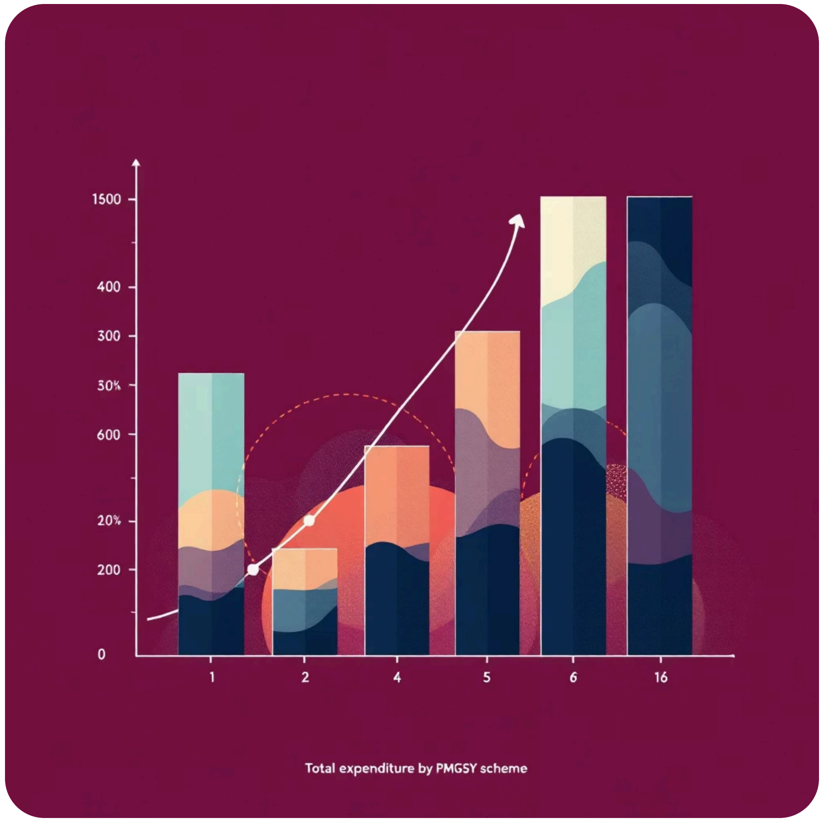
### Action

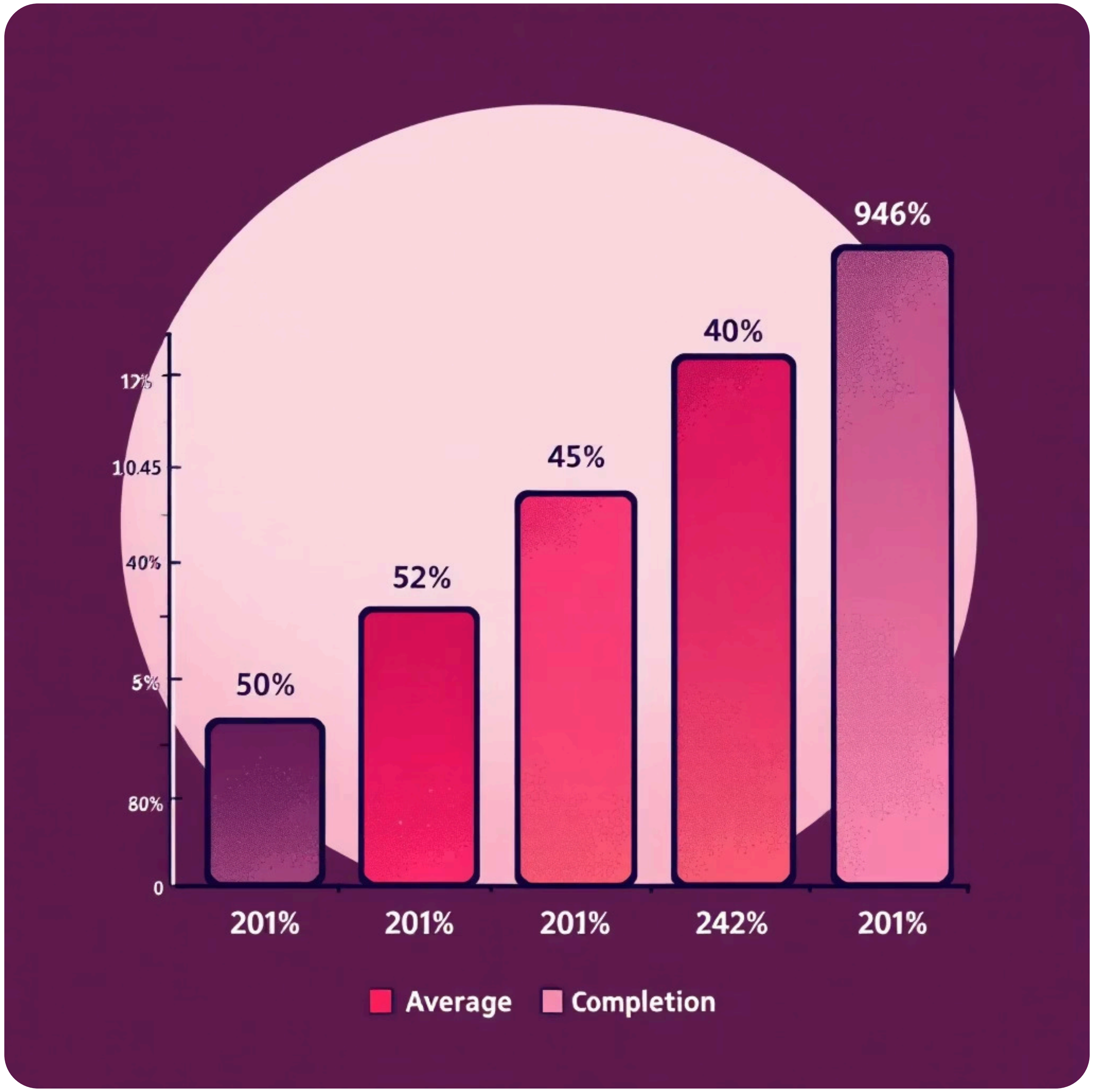
Investigate low-ratio points to understand delays: geography, approvals, or funding timing.



# Scheme-Level Insights (PMGSY Phases)

Aggregate summaries by PMGSY_SCHEME reveal differing maturity and resource allocation. PMGSY-I and II show highest total expenditures and completion percentages (often near/above 90%), while PMGSY-III, RCPLWEA, and PM-JANMAN have lower completion rates, reflecting newer or smaller-scale work.

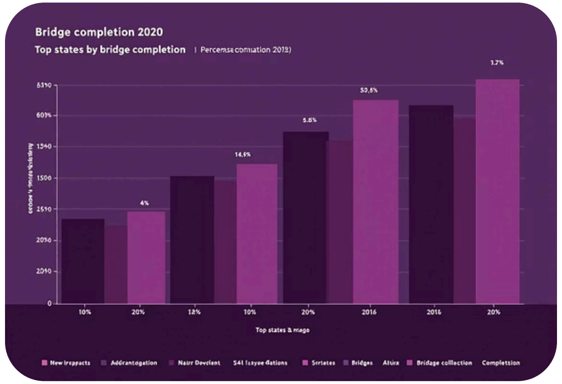



These patterns suggest: older schemes are largely executed, while newer phases need continued monitoring and resources to reach maturity.

# Bridges: Completion & Regional Performance

Bridge completion analysis shows states like Bihar, Odisha, and Andhra Pradesh achieving high average bridge completion percentages (often >85–90%). Bridge metrics help prioritize areas with connectivity gaps over waterways and challenging terrain.




High-performing states provide models for project management, while lower-performing states may face geographic or administrative constraints requiring targeted interventions.

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# Correlation & Feature Relationships

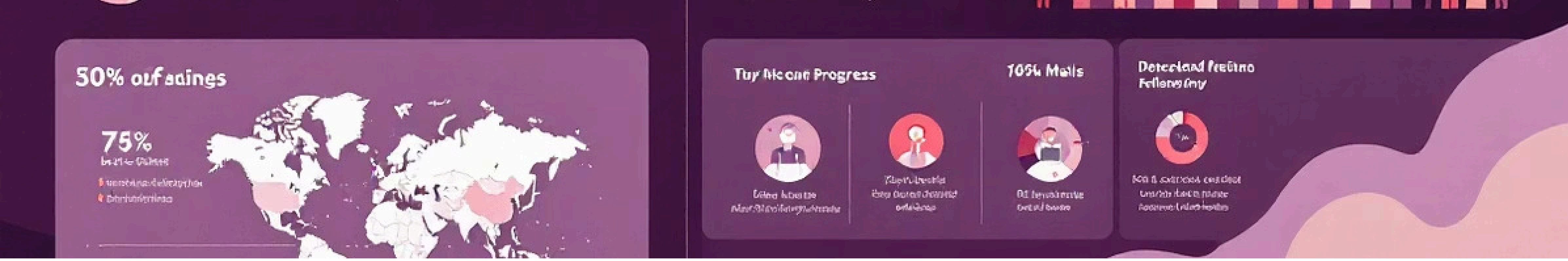
Numeric correlation heatmap (YlGnBu) shows expected relationships: sanctioned counts/lengths correlate positively with completions and costs. Weak correlations indicate independent factors affecting specific outcomes — useful guidance for feature selection in predictive models or prioritization heuristics.

## Practical Takeaways

- Use strong correlations for KPI dashboards (cost ↔️ expenditure, length sanctioned ↔️ completed).
- Investigate weakly correlated variables for root causes (geography, approvals).

## Next Analysis Steps

Time-based trend analysis, district-level hotspot mapping, and predictive modeling for completion risk using enriched geospatial and administrative data.



# Conclusion & Recommendations

This PySpark-driven analysis of PMGSY data demonstrates: concentrated infrastructure activity in large northern/central states, strong cost discipline where projects mature, and high completion in earlier PMGSY phases. Recommendations: focus monitoring on newer schemes (PMGSY-III, RCPLWEA, PM-JANMAN), audit outliers in cost vs expenditure, and deploy targeted support for geographically constrained regions to close connectivity gaps.

### Operational

Prioritize field audits for outliers and accelerate approvals in delayed districts.

### Strategic

Allocate additional resources to newer schemes and replicate best practices from high-performing states.

### Analytical

Build predictive models for project completion risk and a dashboard for real-time monitoring.